

# Complex Analysis

## Problem sheet-2

1. For what values of  $m \in \mathbb{N}$ , does the the function  $f(z) = \frac{z^m}{|z|}$  has a limit as  $z \rightarrow 0$  ?
2. If the limit of a complex valued function at a point exists, then show that the value of the limit is unique.
3. What is the limit of  $\lim_{z \rightarrow 0} \frac{[\operatorname{Re}(z)]^3 + [\operatorname{Im}(z)]^3}{z}$ ?
4. Determine whether or not the function  $f : \mathbb{C} \rightarrow \mathbb{C}$  defined by

$$\forall z \in \mathbb{C}, \quad f(z) = \begin{cases} \frac{\bar{z}^4}{|z|^3} & \text{for } z \neq 0, \\ 0 & \text{for } z = 0 \end{cases}$$

is continuous at  $z = 0$ .

5. Let  $D \subset \mathbb{C}$  be a bounded domain. A function  $f : D \rightarrow \mathbb{C}$  is said to be *Lipschitz* continuous on  $D$  if

$$\exists M > 0 \text{ such that } \forall z_1, z_2 \in D, \quad |f(z_1) - f(z_2)| \leq M|z_1 - z_2|.$$

- a) Which of the functions among  $\operatorname{Re}(z)$ ,  $e^z$ ,  $\sin z$  are Lipschitz continuous on  $D$ ?
  - b) True or false: Lipschitz continuous functions on a domain are uniformly continuous. Justify your answers.
  - c) What about the converse of the statement given in part (b) ?
6. Show that under the stereographic projection, lines in  $\mathbb{C}$  map to circles, through the north pole, on the sphere.
  7. If the point  $P$  on the sphere corresponds to  $z$  under the stereographic projection, show that the *antipodal* <sup>1</sup> point  $-P$  on the sphere corresponds to  $-\frac{1}{\bar{z}}$ .
  8. The stereographic projection  $\sigma : S^2 \rightarrow \hat{\mathbb{C}}$  is given by

$$\sigma(x, y, z) = \frac{x}{1-z} + i \frac{y}{1-z}.$$

Calculate the inverse of  $\sigma$ .

9. Let  $T : \mathbb{C} \rightarrow \mathbb{C}$  be a linear map. Prove that the following statements are equivalent:

<sup>1</sup>Here, *antipodal* means diametrically opposite points on the sphere. If  $P$  has coordinates  $(X, Y, Z)$ , the antipodal point  $-P$  has coordinates  $(-X, -Y, -Z)$ .

- a)  $T$  is uniformly continuous on  $\mathbb{C}$ .
  - b)  $T$  is continuous on  $\mathbb{C}$ .
  - c)  $T$  is continuous at  $0$ .
10. Let  $\overline{\mathbb{D}}$  denote the set  $\{z \in \mathbb{C} : |z| \leq 1\}$ . Prove or disprove that there exists  $z_0 \in \overline{\mathbb{D}}$  such that  $\sup_{z \in \overline{\mathbb{D}}} |\sin(z)| = |\sin(z_0)|$ .
11. Let  $f : D \rightarrow \mathbb{C}$  be a function on a domain  $D \subset \mathbb{C}$ . Show that the following statements are equivalent.
- a)  $f$  is a continuous function on  $D$ ,
  - b) for any open subset  $U$  of  $\mathbb{C}$ ,  $f^{-1}(U)$  is open in  $D$ ,
  - c) for any closed subset  $F$  of  $\mathbb{C}$ ,  $f^{-1}(F)$  is closed in  $D$ .
12. For  $z_1, z_2 \in \mathbb{C}$ , is it true that  $\log(z_1 z_2) = \log(z_1) + \log(z_2)$  always? Elaborate.
13. Let  $z \in \mathbb{C}$  such that  $\text{Arg}(z) \in (-\pi, \pi]$ . Does the limit  $\lim_{z \rightarrow -1} \text{Arg}(z)$  exist? Justify your answers.